



NYU

OSS PULSE

How Coding Agents Reshaped Open-Source Development

Tracking 1.46 billion GitHub events across the 2022–2026 agent era

GROUP 9 · BDAD SPRING 2026

Jhe Chen Li · Bo Yu

ABSTRACT & PLATFORM

What we built · how big · what it found

ABSTRACT

Coding agents are fundamentally reshaping open-source development. We tracked 1.46 B GitHub events across 5 × Q1 snapshots (2022–2026) and found three shifts:

- ① **Development got lighter** — more developers, fewer contributors per repo.
- ② **Weekends disappeared** — activity hit the natural 28.9% baseline in 2026.
- ③ **Two-phase agent adoption** — 2025 explosion → 2026 normalization.

Traditional metrics like stars fail to capture this transformation.

PLATFORM

Apache Spark 3.x

Google Cloud Dataproc (NYU HPC)

SCALE

1.46 B events

930 GB raw → **87.9 GB** cleaned

10 Spark analytics jobs

MOTIVATION

Who needs this · what problem · why now

WHO USES IT

Developers · researchers ·
maintainers · VCs evaluating OSS

WHAT'S BROKEN

Single signals mislead. DeepSeek-
R1 ranks #2 by stars but has no
PyPI library, only 5 push
contributors. Hype ≠ ecosystem.

WHY NOW

The coding-agent era (2024–2026)
is reshaping open-source rhythm —
and stars miss it entirely.

5-YEAR Q1 TIMELINE · AI ECOSYSTEM EVOLUTION

2022

Pre-ChatGPT
baseline

2023

ChatGPT
explosion

2024

LLM bloom
GPT-4 / Claude / Gemini

2025

Year of agents
Cursor / Claude Code /
Devin

2026

Coding-agent
saturation

GOODNESS · WHY OUR RESULTS CAN BE TRUSTED

Five-era data quality validation across 1.46 B events

0

NULLS IN CORE FIELDS

event_type · event_date · actor_login ·
repo_name
across all 5 eras

< 0.39%

2026 RE-VALIDATION DIFF

Re-downloaded 2,160 raw files, re-cleaned,
compared to original output

580

CORRUPT FILES TOLERATED

April 2026 raw .gz files rescued via
spark.sql.files.ignoreCorruptFiles

DATA QUALITY · EVENT VOLUME & FIELD COMPLETENESS BY ERA

Era	Total events	Push events	PR events	null core (4 fields)	Schema
2022-Q1	206.8 M	146.4 M	35.0 M	0	✓ stable
2023-Q1	247.4 M	185.6 M	29.5 M	0	✓ stable
2024-Q1	364.4 M	307.6 M	26.4 M	0	✓ stable
2025-Q1	376.4 M	313.6 M	30.5 M	0	✓ stable
2026-Q1	265.9 M	224.6 M	23.8 M	0	⚠ schema break

2026 schema change details on Slide 9 · full quality report in backup.

DATA SOURCES

GITHUB ARCHIVE

Hourly event logs

Watch · Fork · Push · PR · Issue events from the public GitHub event firehose, archived hourly as compressed JSON.

FORMAT	Hourly .json.gz files
COVERAGE	5 × Q1 snapshots, 2022–2026
VOLUME	1.46 B events · 930 GB raw · 87.9 GB cleaned

HUGGING FACE HUB

Model metadata

Snapshot of every public model on Hugging Face Hub via the huggingface_hub API, including library, downloads, and parameter counts.

FORMAT	JSONL snapshot
COVERAGE	All public models, April 2026
VOLUME	2,815,064 models · 229 MB

PYPI / BIGQUERY

Monthly download counts

Per-library install counts pulled from the public BigQuery PyPI dataset — the closest proxy we have to real engineering adoption.

FORMAT	CSV (project · month · downloads)
COVERAGE	46 AI / ML libraries × 12 months
VOLUME	552 rows · 9 GB BigQuery scan

DATA SAMPLES · GITHUB ARCHIVE · HF HUB · PYPI

Three sources, three schemas — five-row sample of each

GH ARCHIVE

event_type	event_date	actor_login	repo_name	push	pr_merged
WatchEvent	2025-03-15	user_abc	huggingface/transformers	—	—
PushEvent	2025-03-15	dev_xyz	pytorch/pytorch	3	—
PullRequestEvent	2025-03-15	contrib_1	tensorflow/tensorflow	—	true
ForkEvent	2025-03-15	forker_3	openai/whisper	—	—

schema · event_type, event_date, actor_login, repo_name, push_size, push_distinct_size, pr_merged, issue_state, pr_number, payload_action

HUGGING FACE HUB

model_id	library_name	pipeline_tag	downloads	likes
meta-llama/Llama-3-8B	transformers	text-generation	1,234,567	8,920
stabilityai/stable-diffusion	diffusers	image-generation	890,123	5,430
openai/whisper-large-v3	transformers	automatic-speech	456,789	3,210

schema · model_id, author, library_name, pipeline_tag, downloads, likes, created_at, parameter_count, has_safetensors

PYPI / BIGQUERY

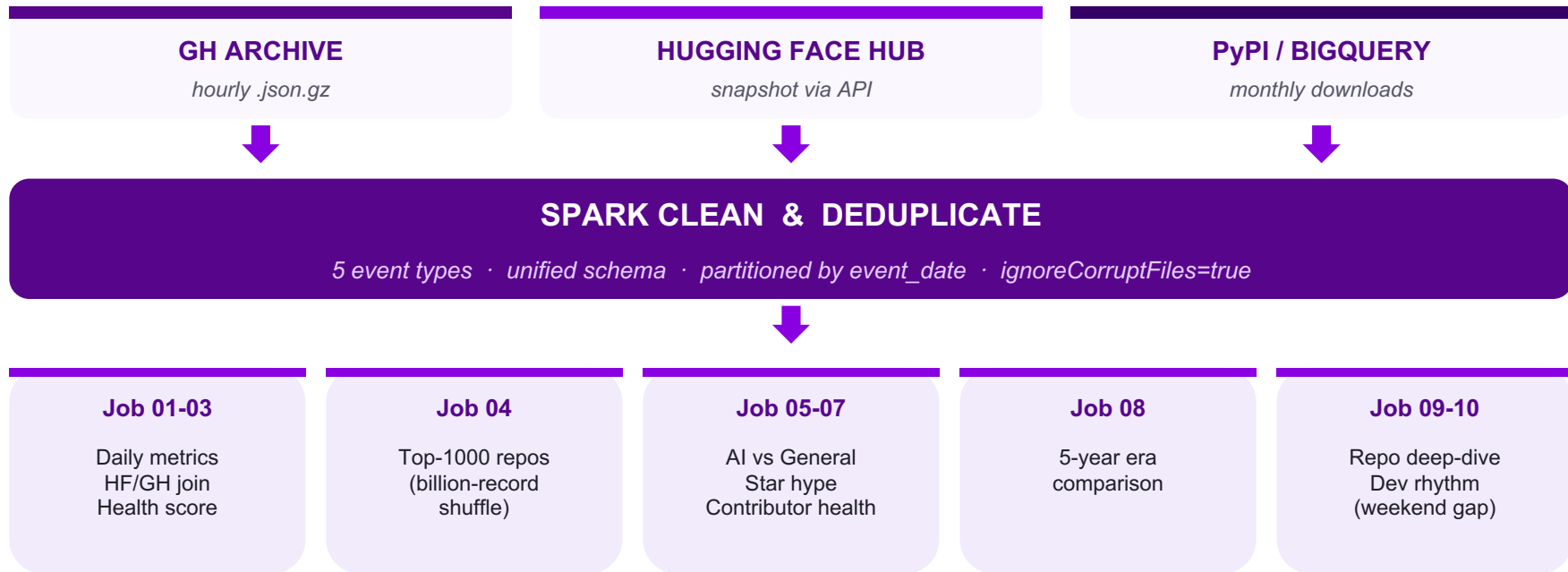
project	month	downloads
transformers	2025-01-01	12,345,678
torch	2025-01-01	45,123,456

schema · project, month (YYYY-MM-01), downloads

Total cleaned scale · 87.9 GB Parquet · 2.81 M HF rows · 552 PyPI rows.

DESIGN DIAGRAM · PIPELINE ARCHITECTURE

3 sources → Spark cleaning → 10 analytics jobs → CSV / Tableau

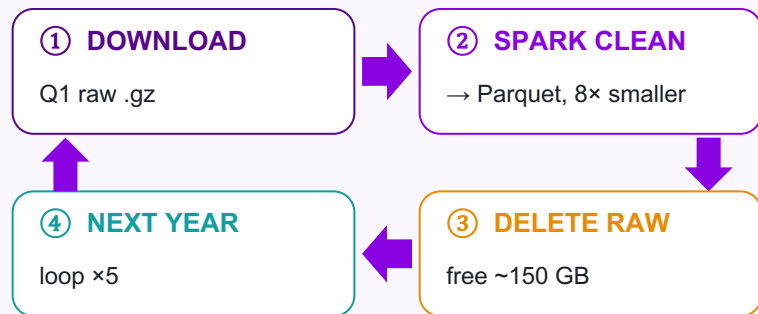


CODE CHALLENGE · Jhe Chen Li

HDFS quota & per-era processing — surviving 5 years × Q1 in 500 GB

ROLLING INGEST · 500 GB QUOTA

Problem · 700 GB needed > 500 GB quota



Outcome · 5-year timeline ingested within 500 GB ✓

PER-ERA PROCESSING · 90 GB ON 48 GB EXECUTORS

Naive cache-all spilled to disk; YARN killed it after 1.5 h. Per-era compute + intermediate saves makes the job crash-safe and resumable.

	Union-first	Per-era
Peak memory	90 GB	15-23 GB
Runtime	2-3 h (killed)	40-60 min
Blast radius	All eras	Only failed era
Shuffle	90 GB / pass	15-23 GB / pass

CODE CHALLENGE · 2026 SCHEMA CHANGE

Job 08 reported merged_prs = 0 for 2026-Q1 — what happened?

THE DISCOVERY

Job 08 reported merged_prs = 0 for 2026-Q1. Investigation revealed the GitHub Archive payload schema changed.

Root cause: GitHub API simplification — announced 2025-08-08, effective 2025-10-07.

Verification: downloaded raw JSON from both eras and compared payload keys directly.

SCHEMA DIFF · 2025 vs 2026

Field	2025	2026
payload.distinct_size	✓ present	X removed
payload.commits	✓ present	X removed
pr.merged	true / false	X NULL
payload.action	opened/closed	+ "merged"

BACKWARD-COMPATIBLE FIX · CODE

```
# Backward-compatible merged-PR detection
((F.col("payload_action") == "closed") & F.col("pr_merged").eqNullSafe(True))
| (F.col("payload_action") == "merged") # 2026+ schema
```

Validation · re-cleaned all 2,160 raw files, diff < 0.39% (corrupt-file tolerance).

CODE CHALLENGE · Bo Yu

Billion-record shuffle — group by repo across 334 date partitions

72 GB

INPUT

83 GB

SHUFFLE

1B+

RECORDS

3+ h

WALL CLOCK

WHAT WE DID

- Tuned `spark.sql.shuffle.partitions` (400 → 200) to cluster size
- `.cache()` on aggregated DataFrame so multi-write actions don't re-shuffle (3× redundancy gone)
- Replaced Python UDFs with native `F.col().isin()`
- Restructured Jobs 05–07 to read pre-aggregated Job 04 output

WHAT WE'D DO DIFFERENTLY

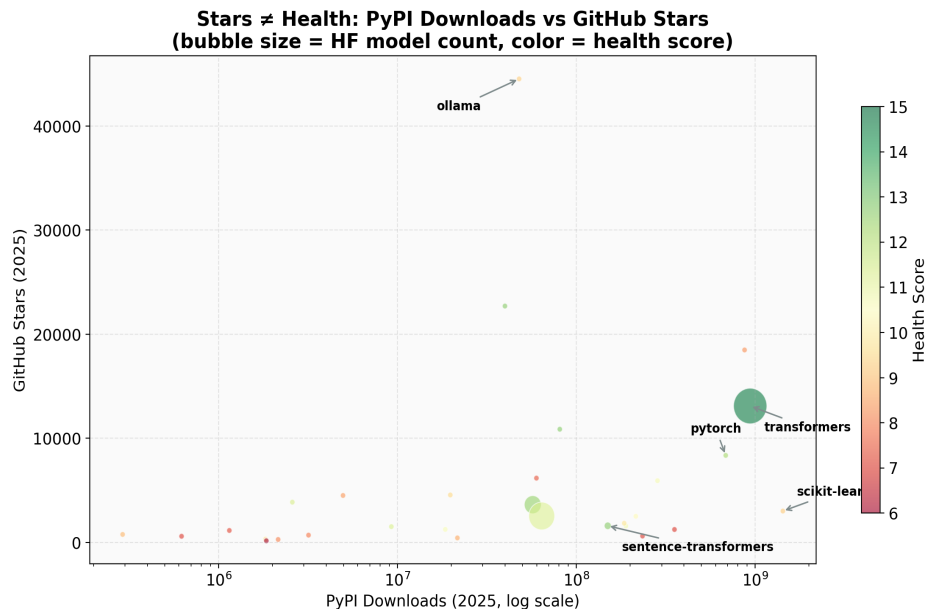
Bucket cleaned data by `repo_name` at ingest:
`cleaned.write`

```
.bucketBy(256, "repo_name")  
.sortBy("repo_name")  
.saveAsTable("gharchive_cleaned")
```

Eliminates the 83 GB shuffle entirely — hours → minutes.

RESULT · STARS ≠ HEALTH

Bubble chart · PyPI downloads × GH stars · size = HF model count



FINDING 1

True leader · transformers 13 K stars · 938 M PyPI
· 835 K HF models — wins on all three signals.

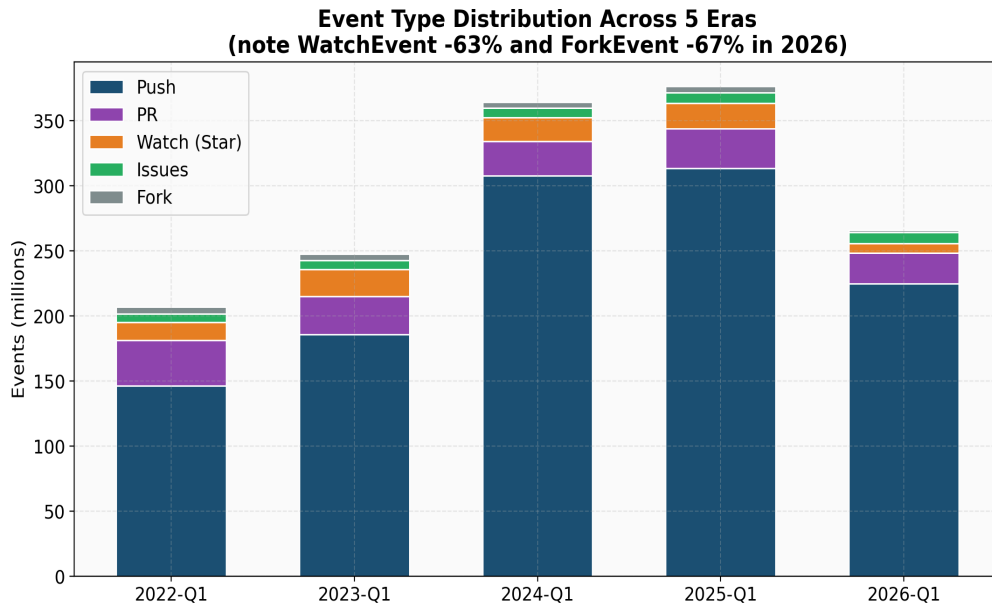
Quiet powerhouse · sentence-transformers 1.6 K
stars → 519 M HF downloads — **327,014×** downloads
/ star.

Asymmetric adoption · ollama 44.5 K stars + 47.9
M PyPI but only 241 HF downloads — it's a runtime, not
a model library.

*“No single signal captures every project —
triangulation is the point.”*

RESULT · EVENT COMPOSITION ACROSS 5 ERAS

Stacked bar · Push · PR · Watch · Issues · Fork by Q1, 2022–2026



FINDING 2

Push holds steady 146 M → 225 M (+54%)

Watch (Star) collapses 19.6 M → 7.2 M (-63%)

Fork collapses 4.6 M → 1.5 M (-67%)

GitHub's removal of inauthentic accounts.

Social signals were hit hardest; push (engineering activity) fell least.

PAPER REFERENCE · EXTERNAL EVIDENCE FOR 2026

Why our 2026 WatchEvent / ForkEvent collapse is consistent with the literature

arXiv > cs > arXiv:2412.13459

Computer Science > Cryptography and Security

[Submitted on 18 Dec 2024 (v1), last revised 6 Sep 2025 (this version, v2)]

Six Million (Suspected) Fake Stars in GitHub: A Growing Spiral of Popularity Contests, Spams, and Malware

Hao He, Haoqin Yang, Philipp Burckhardt, Alexandros Kapravelos, Bogdan Vasilescu, Christian Kästner

~6 M

FAKE STARS IDENTIFIED

Inauthentic stars detected by their classifier

90%

FLAGGED REPOS · observed deleted

Authors re-checked and observed deletion at
16× the normal repo rate

57%

FLAGGED ACCOUNTS · observed deleted

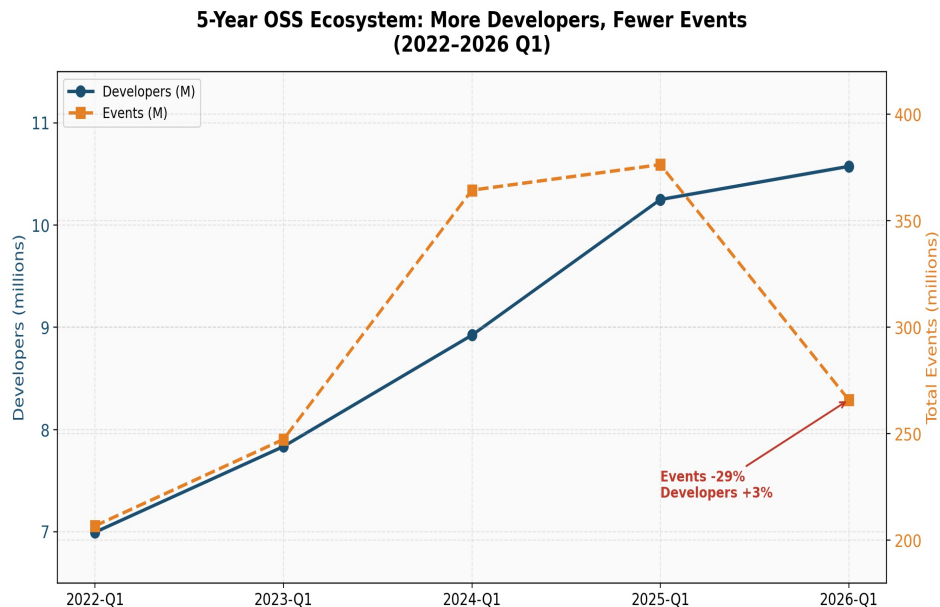
Same re-check on the user accounts that
posted flagged stars

WHY WE CITE · HONESTY ABOUT THE 2026 ANOMALY

Our 2026 **WatchEvent** –63% / **ForkEvent** –67% aligns with their observations. The paper *observes* a 16× anomalous deletion rate and *suggests* active platform enforcement — not an official GitHub announcement.

RESULT · THE 5-YEAR ECOSYSTEM SHIFT

Multi-line chart · developers (M) and total events (M) by Q1, 2022–2026



FINDING 3

Developers 7.0 M → 10.6 M **(+51%)**

Total events 207 M → 266 M **(peaked 376 M)**

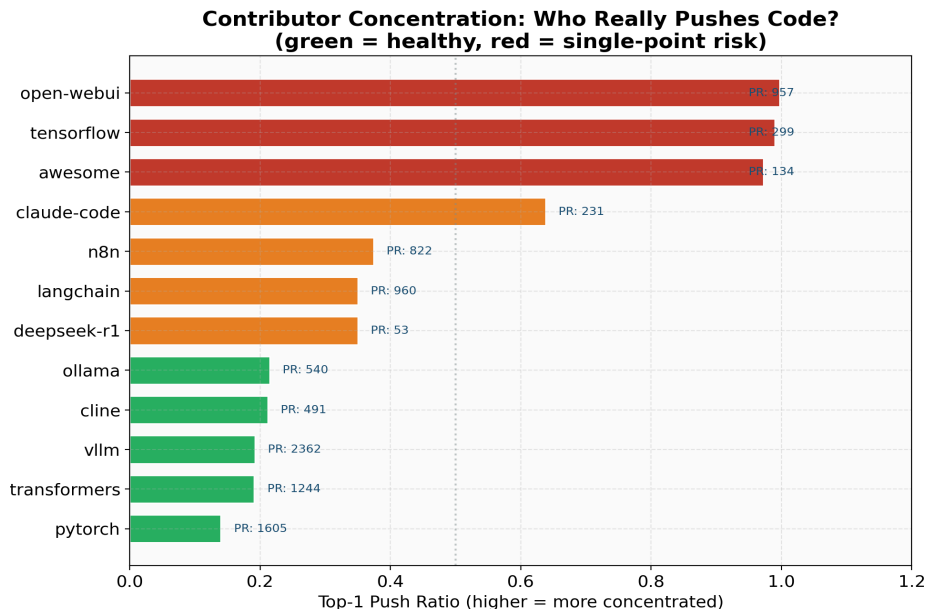
Avg contributors / repo 4.88 → 2.84 **(-42%)**

More developers, more repos — but each repo gets less work and fewer contributors. Development is getting “lighter.”

Caveat · 2026 WatchEvent -63% / ForkEvent -67% align with Hao he. (arXiv:2412.13459): 90% of flagged repos observed deleted (16× normal rate), suggesting platform enforcement.

RESULT · BUS-FACTOR BY REPO

Horizontal bars · top-1 push ratio · color = single-point risk



FINDING 4

Top-1 push ratio across selected AI repos:

Red · > 0.95 open-webui · tensorflow · awesome

Orange · 0.4–0.95 claude-code · n8n · langchain · deepseek-r1

Green · < 0.4 ollama · cline · vllm · transformers · pytorch

Read together with PR contributors —

tensorflow's 0.99 is CI bot automation; metagpt's 0.93 with only 22 PR contributors is real risk.

“Concentration alone doesn't prove fragility.”

BUS-FACTOR · EVIDENCE FROM GITHUB INSIGHTS

The data behind the bars — top-1 contributor dominates these repos

open-webui · top-1 push ratio 0.957

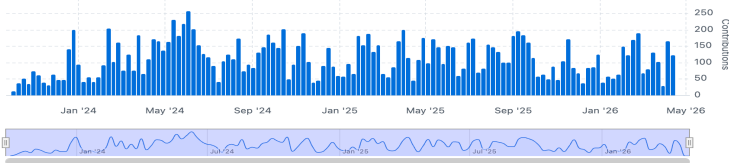
Contributors

Contributions per week to main, line counts have been omitted because commit count exceeds 10,000.

Period: All

Commits over time

Weekly from 2023年9月30日 to 2026年5月2日



tjbeck

11,683 commits

#1

...



Classic298

479 commits

#2

...



#1 tjbeck 11,683 commits · vs **#2 Classic298** 479 commits
= **24x gap**

tensorflow · top-1 push ratio 0.990

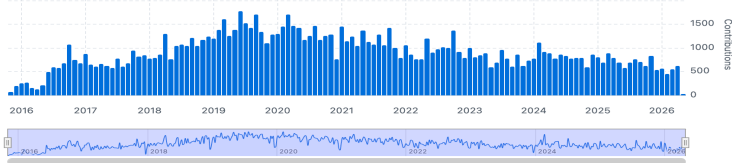
Contributors

Contributions per week to master, line counts have been omitted because commit count exceeds 10,000.

Period: All

Commits over time

Weekly from 2015年11月7日 to 2026年5月2日

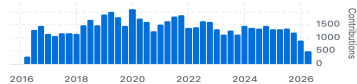


tensorflow-gardener

57,761 commits

#1

...

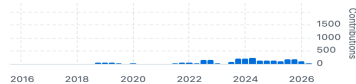


ezhulenev

2,851 commits

#2

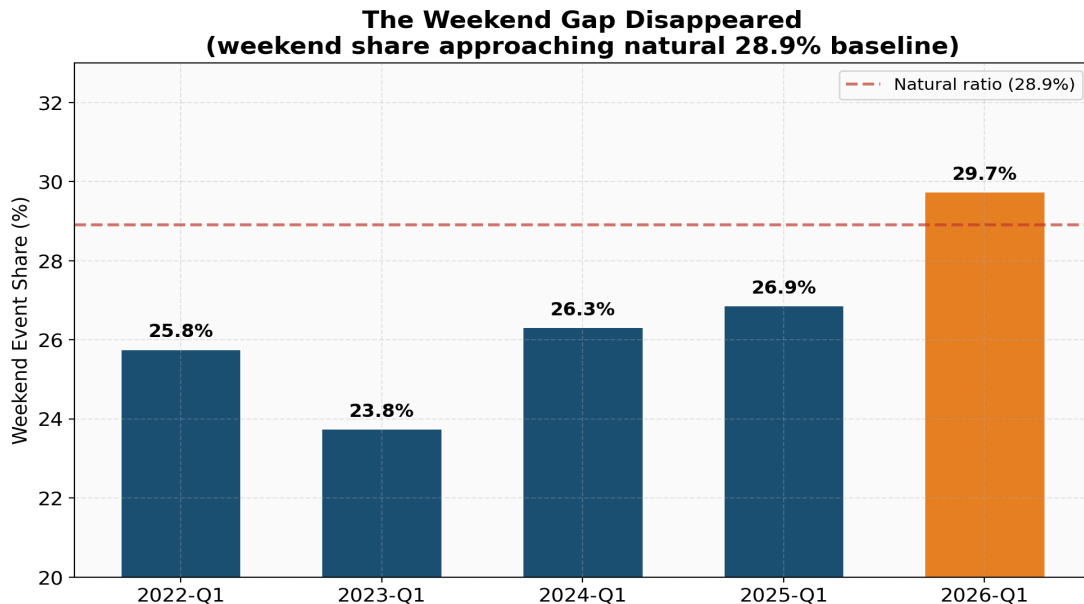
...



#1 tensorflow-gardener 57,761 (CI bot) · **20x #2**

RESULT · THE WEEKEND GAP DISAPPEARED

Bar chart · weekend event share vs 28.9% natural baseline



FINDING 5

Weekend share of all events, by era:

2022 25.8% (−3.1 pp vs natural)

2024 26.3% (−2.6 pp)

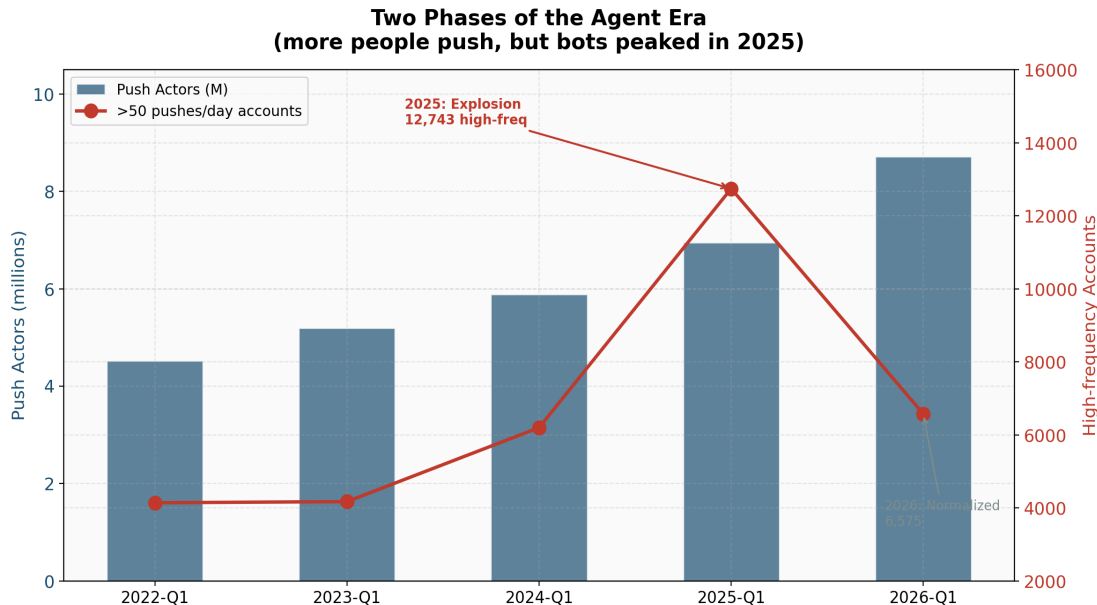
2026 29.7% (+0.8 pp — gap gone)

Push-specific weekend share rose
24.8% → 29.4% — the strongest signal,
since pushes mean writing code.

*“In 2026, GitHub doesn’t know what
day of the week it is anymore.”*

RESULT · TWO PHASES OF THE AGENT ERA

Dual-axis · push actors (M, bars) + >50 pushes/day accounts (line)



FINDING 6

Push actors vs high-frequency accounts:

Phase 1 · 2025 explosion

>50 pushes/day accounts 6,197 → 12,743 (+106%)

Phase 2 · 2026 normalization

Push actors 6.95 M → 8.71 M (+25%)
but >50 pushes/day 12,743 → 6,575 (-48%)

Bots peaked, then GitHub cleaned them. Real broadening of participation took over.

LESSONS LEARNED

What we'd carry forward beyond this project

01

DESIGN FOR THE CONSTRAINT, NOT AGAINST IT

HDFS quota and cross-team permission boundaries forced design decisions that ended up improving the work — rolling pipeline, per-era processing, write-own-read-union.

Carry forward · Treat infrastructure constraints as architectural input, not nuisances. The constraint-driven design was more crash-safe than a quota-free version would have been.

02

NEVER TRUST A LIVE API SCHEMA — ASSERT IT

The 2026 GitHub schema break wasn't loudly announced. Job 08 silently produced `merged_prs = 0` for an entire era before we caught it.

Carry forward · Assert column counts, NULL ratios, and key value ranges explicitly. Compare era-over-era distributions. NULL $\neq$ zero.

03

VALIDATE THE UNEXPECTED WITH OUTSIDE SOURCES

When the 2026 WatchEvent / ForkEvent collapse looked like a bug, cross-checking with He et al. (arXiv 2412.13459) confirmed it as real platform behavior — not a pipeline error.

Carry forward · Anomalies need external corroboration. Triangulating with literature or other data sources turns suspicious numbers into trustworthy ones.

SUMMARY & ACKNOWLEDGEMENTS

The coding-agent era brought three fundamental shifts to OSS

1

DEVELOPMENT GOT LIGHTER

+51% developers but -42% contributors per repo. Single-commit pushes rose 87% → 94%. Pushes became smaller and more frequent.

2

WEEKENDS DISAPPEARED

2026 weekend activity hit 29.7%, finally reaching the natural 28.9% baseline. Coding agents work 24/7.

3

TWO-PHASE AGENT ADOPTION

2025 explosion (>50 pushes/day accounts +106%) → 2026 normalization (broader push base, shorter individual streaks).

IMPLICATION · Traditional metrics (stars, fork counts) are increasingly unreliable in the agent era. Multi-dimensional measurement is essential.

ACKS · NYU HPC — Google Cloud Dataproc · GH Archive · Hugging Face Hub · PyPI / BigQuery.